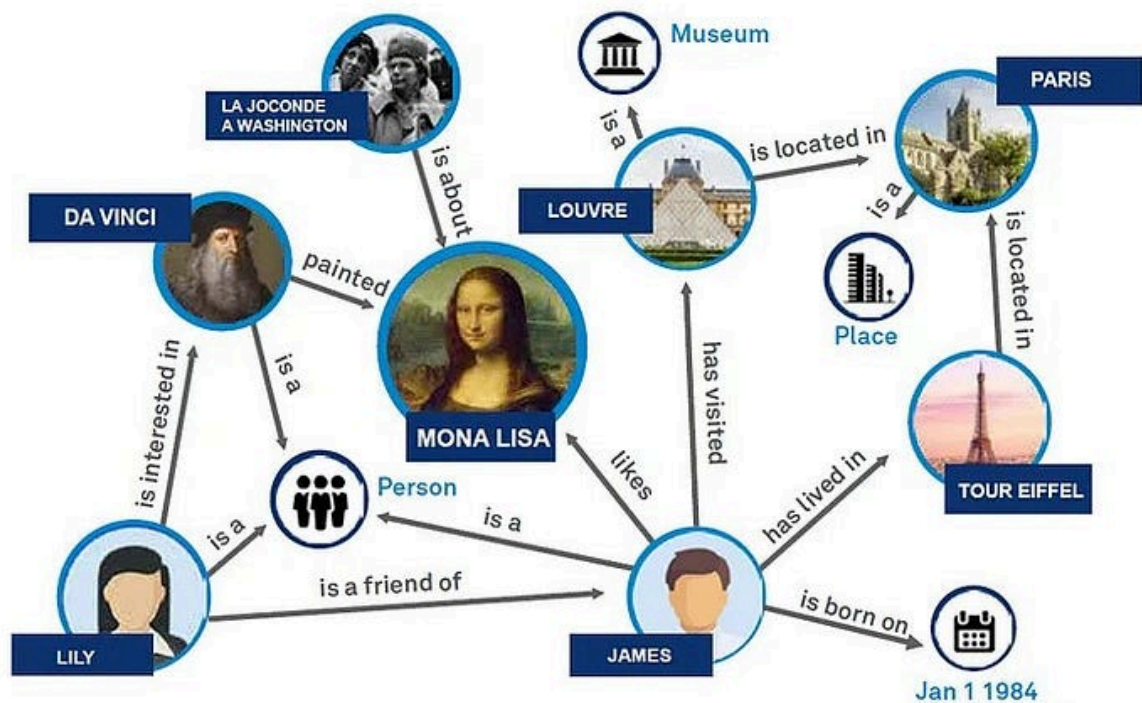


1.Introduction

- Graph Retrieval Augemented Generation **Graph RAG** is gaining momentum and emerging as powerful additional traditional vector search retrieval methods.
- This approach leverages the structured nature of graph databases, which organize data as nodee and relationships, to enhance the depth and cotnexuality retrieved information.



- Graphs are great at representing and storing heterogeneous and interconnected information in a structured manner, effortlessly capturing complex relationships and attributes across diverse data types.
- In contrast, vector databases often struggle with such structured information, as their strength lies in handling unstructured data through high-dimensional vectors.
- In your RAG application, you can combine structured graph data with vector search through unstructured text to achieve the best of both worlds

2. Knowledge graphs are great, but how do you create one?

- Constructing a knowledge graph is typically the most challenging step in leveraging the power of graph-based data representation.
- It involves gathering and structuring the data, which requires a deep understanding of both the domain and graph modeling.
- To simplify this process, we have been experimenting with LLMs. LLMs, with their profound understanding of language and context, can automate significant parts of the knowledge graph creation process.
- By analyzing text data, these models can identify entities, understand the relationships between them, and suggest how they might be best represented in a graph structure.

3. Installation of Packages

```
In [ ]: # %pip install --upgrade --quiet langchain langchain-community langchain-openai
```

4. Import Required Packages

```
In [14]: from langchain_core.runnables import (
    RunnableBranch,
    RunnableLambda,
    RunnableParallel,
    RunnablePassthrough,
)
from langchain_core.prompts import ChatPromptTemplate
from langchain_core.prompts.prompt import PromptTemplate
from langchain_core.pydantic_v1 import BaseModel, Field
from typing import Tuple, List, Optional
from langchain_core.messages import AIMessage, HumanMessage
from langchain_core.output_parsers import StrOutputParser
import os
from langchain_community.graphs import Neo4jGraph
from langchain.document_loaders import WikipediaLoader
from langchain.text_splitter import TokenTextSplitter
from langchain_openai import ChatOpenAI
from langchain_experimental.graph_transformers import LLMGraphTransformer
from neo4j import GraphDatabase
from yfiles_jupyter_graphs import GraphWidget
from langchain_community.vectorstores import Neo4jVector
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores.neo4j_vector import remove_lucene_chars
```

```
from langchain_core.runnables import ConfigurableField, RunnableParallel, Runn
```

5. Configure the Neo4j Environment Setup:

- You need to set up a Neo4j instance follow along with the examples in this blog post. The easiest way is to start a free instance on [Neo4j Aura](#), which offers cloud instances of Neo4j database. Alternatively, you can also set up a local instance of the Neo4j database by downloading the Neo4j Desktop application and creating a local database instance.
- **Note : You can also connect your local database as well I am using Local Neo4j DB to connect.**

```
In [2]: os.environ["OPENAI_API_KEY"] = "sk-"
os.environ["NEO4J_URI"] = "bolt://localhost:7687"
os.environ["NEO4J_USERNAME"] = "neo4j"
os.environ["NEO4J_PASSWORD"] = "admin@123"

graph = Neo4jGraph()
```

6.DataInjection Pipeline

- we will use LLMs I's Wikipedia page. We can utilize LangChain loaders to fetch and split the documents from Wikipedia seamlessly.

```
In [3]: # Read the wikipedia
raw_documents = WikipediaLoader(query="Large Language Model").load()

# Define chunking Strategy
text_splitter = TokenTextSplitter(chunk_size=512, chunk_overlap = 24)
documents = text_splitter.split_documents(raw_documents[:3])
```

```
In [5]: documents
```

Out[5]: [Document(page_content="India, officially the Republic of India (ISO: Bhārat Gaṇarājya), is a country in South Asia. It is the seventh-largest country by area; the most populous country as of June 2023; and from the time of its independence in 1947, the world's most populous democracy. Bounded by the Indian Ocean on the south, the Arabian Sea on the southwest, and the Bay of Bengal on the southeast, it shares land borders with Pakistan to the west; China, Nepal, and Bhutan to the north; and Bangladesh and Myanmar to the east. In the Indian Ocean, India is in the vicinity of Sri Lanka and the Maldives; its Andaman and Nicobar Islands share a maritime border with Thailand, Myanmar, and Indonesia.\nModern humans arrived on the Indian subcontinent from Africa no later than 55,000 years ago.\nTheir long occupation, initially in varying forms of isolation as hunter-gatherers, has made the region highly diverse, second only to Africa in human genetic diversity. Settled life emerged on the subcontinent in the western margins of the Indus river basin 9,000 years ago, evolving gradually into the Indus Valley Civilisation of the third millennium BC E.\nBy 1200 BCE, an archaic form of Sanskrit, an Indo-European language, had diffused into India from the northwest. Its evidence today is found in the hymns of the Rigveda. Preserved by an oral tradition that was resolutely vigilant, the Rigveda records the dawning of Hinduism in India. The Dravidian languages of India were supplanted in the northern and western regions.\nBy 400 BCE, stratification and exclusion by caste had emerged within Hinduism,\nand Buddhism and Jainism had arisen, proclaiming social orders unlinked to heredity.\nEarly political consolidations gave rise to the loose-knit Maurya and Gupta Empires based in the Ganges Basin.\nTheir collective era was suffused with wide-ranging creativity, but also marked by the declining status of women, and the incorporation of untouchability into an organised system of belief. In South India, the Middle kingdoms exported Dravidian-languages scripts and religious cultures to the kingdoms of Southeast Asia.In the early medieval era, Christianity, Islam, Judaism, and Zoroastrianism became established on India's southern and western coasts.\nMuslim armies from Central Asia intermittently overran India's northern plains,\neventually founding the Delhi Sultanate", metadata={'title': 'India', 'summary': "India, officially the Republic of India (ISO: Bhārat Gaṇarājya), is a country in South Asia. It is the seventh-largest country by area; the most populous country as of June 2023; and from the time of its independence in 1947, the world's most populous democracy. Bounded by the Indian Ocean on the south, the Arabian Sea on the southwest, and the Bay of Bengal on the southeast, it shares land borders with Pakistan to the west; China, Nepal, and Bhutan to the north; and Bangladesh and Myanmar to the east. In the Indian Ocean, India is in the vicinity of Sri Lanka and the Maldives; its Andaman and Nicobar Islands share a maritime border with Thailand, Myanmar, and Indonesia.\nModern humans arrived on the Indian subcontinent from Africa no later than 55,000 years ago.\nTheir long occupation, initially in varying forms of isolation as hunter-gatherers, has made the region highly diverse, second only to Africa in human genetic diversity. Settled life emerged on the subcontinent in the western margins of the Indus river basin 9,000 years ago, evolving gradually into the Indus Valley Civilisation of the third millennium BCE.\nBy 1200 BCE, an archaic form of Sanskrit, an Indo-European language, had diffused into India from the northwest. Its evidence today is found in the hymns of the Rigveda. Preserved by an oral tradition that was resolutely vigilant, the Rigveda records the dawning of Hinduism in India. The Dravidian languages of India were supplanted in the northern and western regions.\nBy 400 BCE, stratification and exclusion by caste had emerged within Hinduism,\nand Buddhism and Jainism had arisen, proclaiming social orders unlinked to heredity.\nEarly political consolidations gave rise to the loose-knit

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History

For a continuous duration of nearly 1700 years from the year 1 CE, India was the world's largest economy, constituting 35 to 40% of the world GDP. The combination of protectionist, import-substitution, Fabian socialism, and social democratic-inspired policies governed India for some time after the end of British rule. The economy was then characterised as Dirigism. It had extensive regulation, protectionism, public ownership of large monopolies, pervasive corruption and slow growth. Since 1991, continuing economic liberalisation, metadata={'title': 'Economy of India', 'summary': "The economy of India has transitioned from a mixed planned economy to a mixed middle-income developing social market economy with notable public sector in strategic sectors. It is the world's fifth-largest economy by nominal GDP and the third-largest by purchasing power parity (PPP); on a per capita income basis, India ranked 139th by GDP (nominal) and 127th by GDP (PPP). From independence in 1947 until 1991, successive governments followed Soviet model and promoted protectionist economic policies, with extensive Sovietization, state intervention, demand-side economics, natural resources, bureaucrat driven enterprises and economic regulation. This is characterised as dirigism, in the form of the Licence Raj. The end of the Cold War and an acute balance of payments crisis in 1991 led to the adoption of a broad economic liberalisation in India and indicative planning. Since the start of the 21st century, annual average GDP growth has been 6% to 7%. The economy of the Indian subcontinent was the largest in the world for most of recorded history up until the onset of colonialism in early 19th century. Nearly 70% of India's GDP is driven by domestic consumption; country remains the world's sixth-largest consumer market. Apart from private consumption, India's GDP is also fueled by government spending, investments, and exports. In 2022, India was the world's 8th-largest importer and the 10th-largest exporter. India has been a member of the World Trade Organization since 1 January 1995. It ranks 63rd on the Ease of doing business index and 40th on the Global Competitiveness Index. With 476 million workers, the Indian labour force is the world's second-largest. India has one of the world's highest number of billionaires and extreme income inequality. During the 2008 global financial crisis, the economy faced a mild slowdown. India endorsed Keynesian policy and initiated stimulus measures (both fiscal and monetary) to boost growth and generate demand. In subsequent years, economic growth revived. The period between 2004 and 2014 is referred to as India's lost decade as India fell behind other BRIC economies. In 2021–22, the foreign direct investment (FDI) in India was \$82 billion. The leading sectors for FDI inflows were the service sector, the computer industry, and the telecom industry. India has free trade agreements with several nations and blocs, including ASEAN, SAFTA, Mercosur, South Korea, Japan, Australia, UAE, and several others which are in effect or under negotiating stage. The service sector makes up more than 50% of GDP and remains the fastest growing sector, while the industrial sector and the agricultural sector employs a majority of the labor force. The Bombay Stock Exchange and National Stock Exchange are some of the world's largest stock exchanges by market capitalisation. India is the world's sixth-largest manufacturer, representing 2.6% of global manufacturing output. Nearly 65%

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Document(page_content="Punjab (; Punjabi: [pən'dʒɑːb]), historically known as Panchanada or Pentapotamia, is a state in northern India. Forming part of the larger Punjab region of the Indian subcontinent, the state is bordered by the Indian states of Himachal Pradesh to the north and northeast, Haryana to the south and southeast, and Rajasthan to the southwest; by the Indian union territories of Chandigarh to the east and Jammu and Kashmir to the north. It shares an international border with Punjab, a province of Pakistan to the west. The state covers an area of 50,362 square kilometres (19,445 square miles), which is 1.53% of India's total geographical area, making it the 19th-largest Indian state by area out of 28 Indian states (20th largest, if Union Territories are considered). With over 27 million inhabitants, Punjab is the 16th-largest Indian state by population, comprising 23 districts. Punjabi, written in the Gurmukhi script, is the most widely spoken and the official language of the state. The main ethnic group are the Punjabis, with Sikhs (57.7%) and Hindus (38.5%) forming the dominant religious groups. The state capital, Chandigarh, is a union territory and also the capital of the neighbouring state of Haryana. Three tributaries of the Indus River – the Sutlej, Beas, and Ravi – flow through Punjab. The history of Punjab has witnessed the migration and settlement of different tribes of people with different cultures and ideas, forming a civilisational melting pot. The ancient Indus Valley civilisation flourished in the region until its decline around 1900 BCE. Punjab was enriched during the height of the Vedic period, but declined in predominance with the rise of the Mahajanapadas. The region formed the frontier of initial empires during antiquity including Alexander's and the Maurya empires. It was subsequently conquered by the Kushan Empire, Gupta Empire, and then Harsha's Empire. Punjab continued to be settled by nomadic people; including the Huna, Turkic and the Mongols. Punjab came under Muslim rule c. 1000 CE, and was part of the Delhi Sultanate and the Mughal Empire. Sikhism, based on the teachings of Sikh Gurus, emerged between the 15th and 17th centuries. Conflicts between the", metadata={'title': 'Punjab, India', 'summary': "Punjab (; Punjabi: [pən'dʒɑːb]), historically known as Panchanada or Pentapotamia, is a state in northern India. Forming part of the larger Punjab region of the Indian subcontinent, the state is bordered by the Indian states of Himachal Pradesh to the north and northeast, Haryana to the south and southeast, and Rajasthan to the southwest; by the Indian union territories of Chandigarh to the east and Jammu and Kashmir to the north. It shares an international border with Punjab, a province of Pakistan to the west. The state covers an area of 50,362 square kilometres (19,445 square miles), which is 1.53% of India's total geographical area, making it the 19th-largest Indian state by area out of 28 Indian states (20th largest, if Union Territories are considered). With over 27 million inhabitants, Punjab is the 16th-largest Indian state by population, comprising 23 districts. Punjabi, written in the Gurmukhi script, is the most widely spoken and the official language of the state. The main ethnic group are the Punjabis, with Sikhs (57.7%) and Hindus (38.5%) forming the dominant religious groups. The state capital, Chandigarh, is a union territory and also the capital of the neighbouring state of Haryana. Three tributaries of the Indus River

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t ranking among Indian states in human development index. Punjab has bustling tourism, music, culinary, and film industries.

Etymology

History

Ancient period

The Punjab region is noted as the site of one of the earliest urban societies, the Indus Valley Civilization that flourished from about 3000 B.C. and declined rapidly 1,000 years later, followed by the

metadata={'title': 'Punjab, India', 'summary': "Punjab (; Punjabi: [pən'dʒɑːb]), historically known as Panchanada or Pentapotamia, is a state in northern India. Forming part of the larger Punjab region of the Indian subcontinent, the state is bordered by the Indian states of Himachal Pradesh to the north and northeast, Haryana to the south and southeast, and Rajasthan to the southwest; by the Indian union territories of Chandigarh to the east and Jammu and Kashmir to the north. It shares an international border with Pakistan to the west. The state covers an area of 50,362 square kilometres (19,445 square miles), which is 1.53% of India's total geographical area, making it the 19th-largest Indian state by area out of 28 Indian states (20th largest, if Union Territories are considered). With over 27 million inhabitants, Punjab is the 16th-largest Indian state by population, comprising 23 districts. Punjabi, written in the Gurmukhi script, is the most widely spoken and the official language of the state. The main ethnic group are the Punjabis, with Sikhs (57.7%) and Hindus (38.5%) forming the dominant religious groups. The state capital, Chandigarh, is a union territory and also the capital of the neighbouring state of Haryana. Three tributaries of the Indus River – the Sutlej, Beas, and Ravi – flow through Punjab. The history of Punjab has witnessed the migration and settlement of different tribes of people with different cultures and ideas, forming a civilisational melting pot. The ancient Indus Valley civilisation flourished in the region until its decline around 1900 BC E. Punjab was enriched during the height of the Vedic period, but declined in predominance with the rise of the Mahajanapadas. The region formed the frontier of initial empires during antiquity including Alexander's and the Maurya empires. It was subsequently conquered by the Kushan Empire, Gupta Empire, and then Harsha's Empire. Punjab continued to be settled by nomadic people; including the Huna, Turkic and the Mongols. Punjab came under Muslim rule c. 1000 CE, and was part of the Delhi Sultanate and the Mughal Empire. Sikhism, based on the teachings of Sikh Gurus, emerged between the 15th and 17th centuries. Conflicts between the Mughals and the later Sikh Gurus precipitated a militarisation of the Sikhs, resulting in the formation of a confederacy after the weakening of the Mughal Empire, which competed for control with the larger Durrani Empire. This confederacy was united in 1801 by Maharaja Ranjit Singh, forming the Sikh Empire. The larger Punjab region was annexed by the British East India Company from the Sikh Empire in 1849. At the time of the independence of India from British rule in 1947, the Punjab province was partitioned along religious lines amidst widespread violence, with the Muslim-majority western portion becoming part of Pakistan and the Hindu- and Sikh-majority east remaining in India, causing a large-scale migration between the two. After the Punjabi Suba movement, Indian Punjab was reorganised on the basis of language in 1966, when its Haryanvi- and Hindi-speaking areas were carved out as Haryana, Pahari-speaking regions attached to Himachal Pradesh and the remaining, mostly Punjabi-speaking areas became the current state of Punjab. A separatist insurgency occurred in the state during the 1980s. At present, the economy of Punjab is the 15th-largest state economy in India with ₹5.29 trillion (US\$66 billion) in gross domestic product and a per capita GDP of ₹151,367 (US\$1,900), ranking 17th among Indian states. Since independence, Punjab is predominantly an agrarian society. It is the ninth-highest ranking among Indian states in human development index. Punjab has bustling tourism, music, culina

```
ry, and film industries.", 'source': 'https://en.wikipedia.org/wiki/Punjab,_India'}})]
```

Now it's time to construct a graph based on the retrieved documents. For this purpose, we have implemented an `LLMGraphTransformer` module that significantly simplifies constructing and storing a knowledge graph in a graph database.

```
In [7]: llm = ChatOpenAI(temperature=0,model_name = "gpt-3.5-turbo-0125")
llm_transformer = LLMGraphTransformer(llm=llm)

graph_documents = llm_transformer.convert_to_graph_documents(documents)
graph.add_graph_documents(graph_documents, baseEntityLabel=True, include_source=True)
```

- **LLM Selection and Support:**

- You can define which LLM you want the knowledge graph generation chain to use.
- Currently, we support only function calling models from OpenAI and Mistral.
- Expansion Plans: We plan to expand the LLM selection in the future.
- Example Model: In this example, we are using the latest GPT-4.
- Quality Dependency: Note that the quality of generated graph significantly depends on the model you are using.
- Optimal Model Usage: In theory, you always want to use the most capable one.

- **Graph Generation Process:**

- The LLM graph transformers return graph documents.
- Integration with Neo4j: These documents can be imported to Neo4j via the `add_graph_documents` method.
- BaseEntityLabel Parameter: It assigns an additional **Entity** label to each node, enhancing indexing and query performance.
- Include Source Parameter: It links nodes to their originating documents, facilitating data traceability and context understanding.

- **Graph Inspection:**

- Visualization Tool: You can inspect the generated graph with `yFiles` visualization.

```
In [8]: # directly show the graph resulting from the given cypher query
default_cypher = "MATCH (s)-[r:!MENTIONS]->(t) RETURN s,r,t LIMIT 50"
```

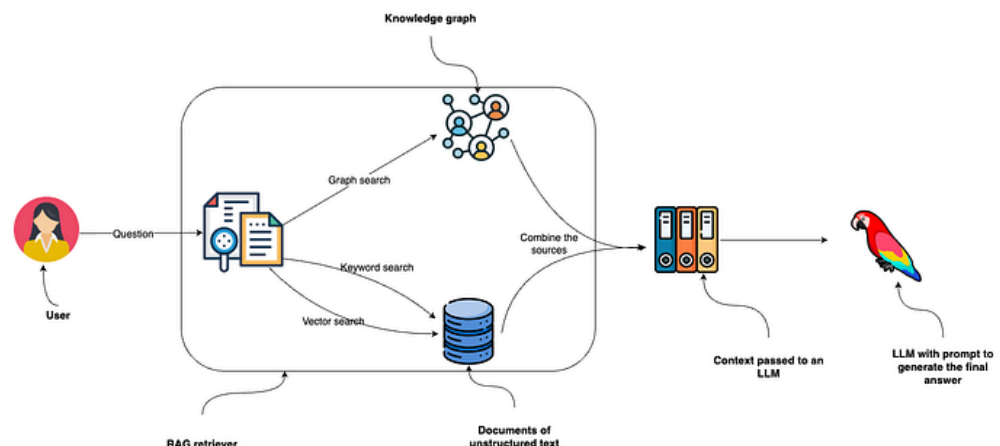
```
def showGraph(cypher:str = default_cypher):
    # create a neo4j session to run queries
    driver = GraphDatabase.driver(
        uri = os.environ["NEO4J_URI"],
        auth = (os.environ["NEO4J_USERNAME"],os.environ["NEO4J_PASSWORD"]))
    session = driver.session()
    widget = GraphWidget(graph = session.run(cypher).graph())
    widget.node_label_mapping = "id"
    # display(widget)
    return widget

showGraph()
```

GraphWidget(layout=Layout(height='800px', width='100%'))

7. Hybrid Retrieval for RAG:

- After graph generation, a hybrid retrieval approach is utilized for RAG (Retrieval-Augmented Generation) applications.



- **Retrieval Process:**

- **User Interaction:** The process begins with a user posing a question.
- **RAG Retriever:** The question is directed to an RAG retriever.
- **Retrieval Techniques:** This retriever employs keyword and vector searches to sift through unstructured text data.
- **Graph Integration:** It combines retrieved information with data from the knowledge graph.
- **Single Database System:** Since Neo4j supports both keyword and vector indexes, all three retrieval options can be implemented using a single database system.
- **Final Answer Generation:** The collected data from these

sources is fed into an LLM to generate and deliver the final answer.

- **Unstructured Data Retriever:**

- **Neo4jVector.from_existing_graph Method:**

- **Functionality:** You can utilize this method to add both keyword and vector retrieval to documents.
 - **Configuration:** It configures keyword and vector search indexes for a hybrid search approach.
 - **Target Nodes:** This method targets nodes labeled Document.
 - **Additional Functionality:** It calculates text embedding values if they are missing.

```
In [9]: vector_index = Neo4jVector.from_existing_graph(  
    OpenAIEmbeddings(),  
    search_type="hybrid",  
    node_label="Document",  
    text_node_properties=["text"],  
    embedding_node_property="embedding"  
)
```

- **Vector Index Utilization:**

- **Similarity_Search Method:**

- **Usage:** The vector index can be called with the similarity_search method.

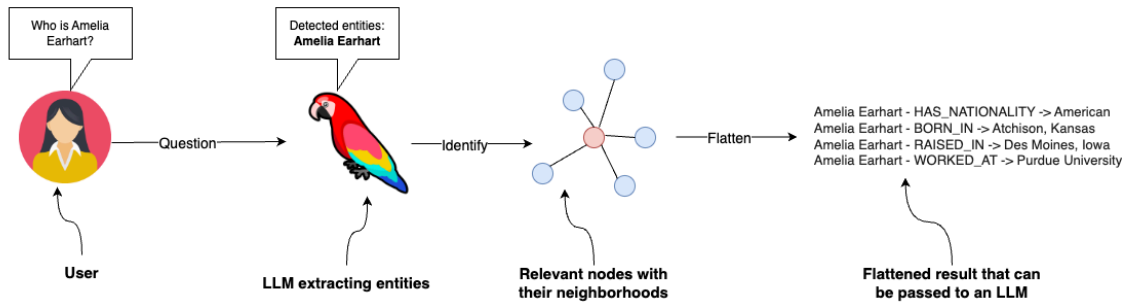
8.Graph Retriever:

- **Configuration and Process:**

- **Involvement and Flexibility:** Configuring a graph retrieval is more involved but offers more freedom.
 - **Methodology:** It utilizes a full-text index to identify relevant nodes and then returns their direct neighborhood.

- **Example and Visualization:**

- **Example Diagram:**



- **Retriever Workflow:**

- Identification of Relevant Entities: The graph retriever starts by identifying relevant entities in the input.
- Structured Output Instruction: For simplicity, the LLM is instructed to identify people, organizations, and locations.

- **Utilization of LCEL:**

- Method: To achieve this, LCEL is used with the newly added `with_structured_output` method.

```

In [15]: # Retriever
graph.query(
    "CREATE FULLTEXT INDEX entity IF NOT EXISTS FOR (e:__Entity__) ON EACH [e.

# Extract entities from text
class Entities(BaseModel):
    """Identifying information about entities."""

    names: List[str] = Field(
        ...,
        description="All the person, organization, or business entities that "
        "appear in the text",
    )

prompt = ChatPromptTemplate.from_messages(
    [
        ("system", "You are extracting organization and person entities from t
        ("human", "Use the given format to extract information from the followi
    ]
)

entity_chain = prompt | llm.with_structured_output(Entities)
  
```

```

c:\Users\ibm26\anaconda3\envs\dev-llm\lib\site-packages\langchain_core\_api\bet
a_decorator.py:86: LangChainBetaWarning: The function `with_structured_output`
is in beta. It is actively being worked on, so the API may change.
warn_beta(
  
```

```

In [18]: entity_chain.invoke({"question": "India is located in south asia?"}).names
  
```

```

Out[18]: ['India', 'south asia']
  
```


9.Full Knowledge Graph

Great, now that we can detect entities in the question, let's use a full-text index to map them to the knowledge graph. First, we need to define a full-text index and a function that will generate full-text queries that allow a bit of misspelling, which we won't go into much detail here.

```
In [21]: def generate_full_text_query(input: str) -> str:
        """
        Generate a full-text search query for a given input string.

        This function constructs a query string suitable for a full-text search.
        It processes the input string by splitting it into words and appending a
        similarity threshold (~2 changed characters) to each word, then combines
        them using the AND operator. Useful for mapping entities from user questions
        to database values, and allows for some misspellings.
        """
        full_text_query = ""
        words = [el for el in remove_lucene_chars(input).split() if el]
        for word in words[:-1]:
            full_text_query += f" {word}~2 AND"
        full_text_query += f" {words[-1]}~2"
        return full_text_query.strip()

# Fulltext index query
def structured_retriever(question: str) -> str:
    """
    Collects the neighborhood of entities mentioned
    in the question
    """
    result = ""
    entities = entity_chain.invoke({"question": question})
    for entity in entities.names:
        response = graph.query(
            """CALL db.index.fulltext.queryNodes('entity', $query, {limit:2})
            YIELD node,score
            CALL {
                MATCH (node)-[r:!MENTIONS]->(neighbor)
                RETURN node.id + ' - ' + type(r) + ' -> ' + neighbor.id AS output
            UNION
            MATCH (node)<-[r:!MENTIONS]-(neighbor)
            RETURN neighbor.id + ' - ' + type(r) + ' -> ' + node.id AS output
            }
            RETURN output LIMIT 50
            """,
            {"query": generate_full_text_query(entity)},
        )
        result += "\n".join([el['output'] for el in response])
    return result
```

```
In [24]: print(structured_retriever("India is located in south asia?"))
```

Large Language Model - TASK -> General-Purpose Language Generation
 Large Language Model - TASK -> Natural Language Processing
 Large Language Model - TASK -> Classification
 Large Language Model - MODEL -> Artificial Neural Networks
 Large Language Model - MODEL -> Decoder-Only Transformer-Based Architecture
 Large Language Model - MODEL -> Recurrent Neural Network Variants
 Large Language Model - MODEL -> Mamba
 Large Language Model - METHOD -> Fine Tuning
 Large Language Model - MODEL -> Gpt-3
 Large Language Model - COMPONENT -> Syntax
 Large Language Model - COMPONENT -> Semantics
 Large Language Model - COMPONENT -> Ontology
 Large Language Model - COMPONENT -> Inaccuracies
 Large Language Model - COMPONENT -> Biases
 Large Language Model - COMPANY -> Openai
 Large Language Model - MODEL -> Chatgpt
 Large Language Model - MODEL -> Microsoft Copilot
 Large Language Model - COMPANY -> Google
 Large Language Model - MODEL -> Palm
 Large Language Model - MODEL -> Gemini
 Neurips Conference - CONFERENCE -> Google Researchers
 Google Researchers - ARCHITECTURE -> Transformer Architecture
 Transformer Architecture - COMPONENT -> Attention Mechanism
 Attention Mechanism - RESEARCHER -> Bahdanau
 Bahdanau - TECHNOLOGY -> Seq2Seq Technology
 Seq2Seq Technology - MODEL -> Bert
 Large Language Model - MODEL -> Gpt-1
 Large Language Model - MODEL -> Gpt-2
 Gpt-3 - TECHNOLOGY -> Api
 Chatgpt - RELEASED_IN -> 2022
 Chatgpt - DEPLOYED_ON -> Browser-Based
 Chatgpt - TARGETED_TOWARDS -> General Population
 Chatgpt - CAUSED -> Media Hype
 Gpt-4 - RELEASED_IN -> 2023
 Gpt-4 - PRAISED_FOR -> Increased Accuracy
 Gpt-4 - PRAISED_FOR -> Multimodal Capabilities
 Gpt-4 - DESCRIBED_AS -> Holy Grail
 Openai - WITHHELD -> Did Not Reveal Architecture
 Openai - WITHHELD -> Did Not Reveal Number Of Parameters
 Gpt-4 - MENTIONED_WITH -> Competing Language Models
 Gpt-4 - OBSERVED_IN -> Playing Catch-Up
 Google - DEVELOPED -> T5-11B
 Google - DEVELOPED -> Palm-E
 Google - DEVELOPED -> Bard (Gemini Pro)
 T5-11B - RELEASED_IN -> 2019
 Palm-E - RELEASED_IN -> 2022
 Claude 3 - RELEASED_IN -> 2024
 Bard (Gemini Pro) - SURPASSED -> January 26, 2024
 Bard (Gemini Pro) - SURPASSED -> Gpt-4
 Gpt-4 - NOT_SURPASSED -> Gpt-4-Turbo
 Large Language Model - TASK -> General-Purpose Language Generation
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10.Final retriever

As we mentioned at the start, we'll combine the unstructured and graph retriever to create the final context that will be passed to an LLM.

```
In [26]: def retriever(question: str):
    print(f"Search query: {question}")
    structured_data = structured_retriever(question)
    unstructured_data = [el.page_content for el in vector_index.similarity_search(question)]
    final_data = f"""Structured data:
    {structured_data}
    Unstructured data:
    {"#Document ".join(unstructured_data)}
    """
    return final_data
```

As we are dealing with Python, we can simply concatenate the outputs using the f-string.

11. Defining the RAG chain

We have successfully implemented the retrieval component of the RAG. First, we will introduce the query rewriting part that allows conversational follow up questions.

```
In [27]: # Condense a chat history and follow-up question into a standalone question
_template = """Given the following conversation and a follow up question, rephrase the question into its original language.
Chat History:
{chat_history}
Follow Up Input: {question}
Standalone question: """ # noqa: E501
CONDENSE_QUESTION_PROMPT = PromptTemplate.from_template(_template)

def _format_chat_history(chat_history: List[Tuple[str, str]]) -> List:
    buffer = []
    for human, ai in chat_history:
        buffer.append(HumanMessage(content=human))
        buffer.append(AIMessage(content=ai))
    return buffer

_search_query = RunnableBranch(
    # If input includes chat_history, we condense it with the follow-up question
    (
        RunnableLambda(lambda x: bool(x.get("chat_history"))).with_config(
            run_name="HasChatHistoryCheck"
        ), # Condense follow-up question and chat into a standalone question
        RunnablePassthrough.assign(
            chat_history=lambda x: _format_chat_history(x["chat_history"])
        )
    ), CONDENSE_QUESTION_PROMPT
    ChatOpenAI(temperature=0)
    StrOutputParser(),
),
```

```

    # Else, we have no chat history, so just pass through the question
    RunnableLambda(lambda x : x["question"]),
)

```

```

In [28]: template = """Answer the question based only on the following context:
{context}

Question: {question}
Use natural language and be concise.
Answer: """
prompt = ChatPromptTemplate.from_template(template)

chain = (
    RunnableParallel(
        {
            "context": _search_query | retriever,
            "question": RunnablePassthrough(),
        }
    )
    | prompt
    | llm
    | StrOutputParser()
)

```

```

In [29]: chain.invoke({"question": "What is large language model?"})

```

Search query: What is large language model?

```

Out[29]: 'A large language model is an artificial neural network used for general-purpose language generation, natural language processing, and classification tasks. It learns statistical relationships from text documents during training to generate text and understand syntax, semantics, ontology, inaccuracies, and biases in human language.'

```

```

In [30]: chain.invoke(
    {
        "question": "What is the name of first large language model?",
        "chat_history": [("What is large language model?", "A large language model")]
    }
)

```

Search query: What is the name of the first large language model?

```

Out[30]: 'The name of the first large language model is GPT-1.'

```